

# TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

## PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 07

**75 minutes**

### INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

**Calculators and dictionaries are NOT permitted.**

**There is no formulae booklet for this test.**

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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**1** Which of the following functions is undefined at exactly two real values of  $x$ ?

**A**  $f(x) = \frac{1}{(x-2)^2}$

**B**  $f(x) = \frac{1}{x^2}$

**C**  $f(x) = \sqrt{x-2}$

**D**  $f(x) = \frac{1}{x^2 + 4}$

**E**  $f(x) = \frac{1}{x^2 - 4}$

- 2 Find  $\int \left(x + \frac{1}{x}\right)^2 dx$ .
- A**  $\frac{x^3}{3} + 2x - \frac{1}{x} + c$
- B**  $\frac{3}{x^3} - \frac{1}{x} + c$
- C**  $\frac{1}{3^3} \left(x + \frac{1}{x}\right)^3 + c$
- D**  $\frac{x^3}{3} + 2x + \frac{1}{x} + c$
- E**  $\frac{3}{x^3} + 2x + c$
- F**  $x^3 + 2x - \frac{1}{x} + c$

- 3** An arithmetic series has first term 5 and common difference 3. Find the sum of the first 40 terms.
- A** 2440
  - B** 2540
  - C** 1270
  - D** 2600
  - E** 2500
  - F** 2580

4 A student solves  $x^2 = 4x$ .

I Divide both sides by  $x$ :  $x = 4$ .

II So the only solution is  $x = 4$ .

Which of the following is correct?

A The first error is on line I: dividing by  $x$  discards the solution  $x = 0$ .

B The solution is completely correct.

C The first error is on line II.

D The first error is on line I:  $x^2 \div x = 2x$ .

E The first error is on line I: it should give  $x = 2$ .

- 5  $p(x)$  is a polynomial.  $P$ : ' $p(2) = 0$ '.  $Q$ : ' $(x - 2)^2$  is a factor of  $p(x)$ '. Which one of the following is true?
- A  $P$  is necessary and sufficient for  $Q$ .
  - B  $P$  is necessary but not sufficient for  $Q$ .
  - C  $P$  is sufficient but not necessary for  $Q$ .
  - D  $P$  is neither necessary nor sufficient for  $Q$ .

- 6** In a class every student studies at least one of Maths and Physics. 70% study Maths and 60% study Physics. A student is picked at random. What is the probability they study both?
- A** 0.42
  - B** 0.2
  - C** 0.1
  - D** 0.3
  - E** 0.5
  - F** it cannot be determined

- 7 Consider the statement: **\*\*if  $n$  is a multiple of 6, then  $n$  is a multiple of 3.\*\*** Which of the following is its contrapositive?
- A** If  $n$  is not a multiple of 3, then  $n$  is not a multiple of 6.
  - B** If  $n$  is a multiple of 3, then  $n$  is a multiple of 6.
  - C** If  $n$  is not a multiple of 6, then  $n$  is not a multiple of 3.
  - D**  $n$  is a multiple of 6 if and only if  $n$  is a multiple of 3.
  - E** If  $n$  is a multiple of 3, then  $n$  is not a multiple of 6.

- 8 To prove **if  $n^2$  is even then  $n$  is even** by contradiction, which assumption should be made?
- A  $n$  is odd.
  - B  $n^2$  is even and  $n$  is odd.
  - C  $n^2$  is even.
  - D  $n^2$  is odd.
  - E  $n$  is even and  $n^2$  is odd.

- 9 A statement reads: **\*\*for every positive integer  $n$ , there is a prime  $p$  with  $n < p < 2n$ .\*\*** Which of the following is its negation?
- A For every positive integer  $n$ , there is a prime  $p$  with  $p \leq n$  or  $p \geq 2n$ .
  - B There is a positive integer  $n$  such that no prime  $p$  satisfies  $n < p < 2n$ .
  - C There is a positive integer  $n$  and a prime  $p$  with  $p \leq n$  or  $p \geq 2n$ .
  - D There is a prime  $p$  such that for every positive integer  $n$ ,  $n < p < 2n$  fails.
  - E For every positive integer  $n$ , no prime  $p$  satisfies  $n < p < 2n$ .

**10** A set of five distinct positive integers has median 10 and mean 10. Which of the following statements must be true?

- I** The largest value is at least 12.
- II** The smallest value is at most 8.
- III** The range is at least 4.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 11** A rectangle has integer side lengths and its area (in  $\text{cm}^2$ ) is numerically equal to its perimeter (in cm). How many such rectangles are there (counting  $a \times b$  and  $b \times a$  once)?
- A** 4
  - B** 3
  - C** 0
  - D** 1
  - E** 2
  - F** infinitely many

- 12** Let  $f$  be a polynomial.  $P$ : ' $f'(x) > 0$  for all real  $x$ '.  $Q$ : ' $f(x) = 0$  has exactly one real root'. Which one of the following is true?
- A**  $P$  is necessary and sufficient for  $Q$ .
  - B**  $P$  is necessary but not sufficient for  $Q$ .
  - C**  $P$  is sufficient but not necessary for  $Q$ .
  - D**  $P$  is neither necessary nor sufficient for  $Q$ .

- 13 The expansion of  $\left(2x - \frac{1}{x^2}\right)^9$  contains a term in  $x^k$  for each of which of the following values of  $k$ ?
- I  $k = 0$
  - II  $k = 3$
  - III  $k = -6$
- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

- 14** Five friends each shake hands with some of the others (each pair shakes hands at most once). Which of the following is impossible?
- A** Every friend shakes hands with exactly two others.
  - B** Every friend shakes hands with exactly four others.
  - C** The numbers of handshakes made are 1, 1, 2, 2, 2.
  - D** The numbers of handshakes made by the five friends are 4, 4, 4, 4, 1.
  - E** Exactly one friend shakes no hands at all.

- 15 A teacher says: '**You will pass only if you revise.**' Which of the following statements means the same thing?
- A If you pass, then you revised.
  - B If you revise, then you will pass.
  - C You will pass if you revise.
  - D If you do not pass, then you did not revise.
  - E You will pass if and only if you revise.

- 16** Claim: for all positive real  $x$ ,  $x + \frac{1}{x} \geq 2$ .
- I** Suppose  $x + \frac{1}{x} \geq 2$ .
- II** Multiply by  $x > 0$ :  $x^2 + 1 \geq 2x$ .
- III** So  $x^2 - 2x + 1 \geq 0$ , i.e.  $(x - 1)^2 \geq 0$ , which is true.
- IV** Hence the claim is proved.

Which of the following best describes the argument?

- A** The argument assumes what it is trying to prove (line I); it would be valid if the steps were reversed, starting from  $(x - 1)^2 \geq 0$ .
- B** The argument is completely correct as written.
- C** The first error is on line II: multiplying by  $x$  reverses the inequality.
- D** The first error is on line III.
- E** The claim is false.

**17** Here is a proof that there are infinitely many primes, with the steps out of order.

**I** Then  $N$  leaves remainder 1 on division by each of  $p_1, \dots, p_n$ , so no  $p_i$  divides  $N$ .

**II** Suppose there are only finitely many primes  $p_1, p_2, \dots, p_n$ , and let  $N = p_1 p_2 \cdots p_n + 1$ .

**III** So  $N$  has a prime factor that is not in the list, which is a contradiction.

**IV** But  $N > 1$ , so  $N$  has at least one prime factor.

Which is the correct order?

**A** II, I, IV, III

**B** II, III, I, IV

**C** I, II, IV, III

**D** II, I, III, IV

**E** IV, II, I, III

**18**  $p(x)$  is a polynomial of degree 4 with real coefficients. Which of the following statements must be true?

- I**  $p(x) = 0$  has at least one real root.
- II**  $p(x) = 0$  has at most four real roots.
- III** The graph of  $y = p(x)$  has at least one turning point.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

**19**  $n$  is a positive integer. Which of the following statements is/are true?

**I** The coefficient of  $x^n$  in  $(1+x)^{2n}$  is  $\binom{2n}{n}$ .

**II** The sum of the coefficients in the expansion of  $(1-x)^n$  is 0.

**III**  $(1+x)^n \geq 1+nx$  for all  $x \geq 0$ .

**A** none of them

**B** I only

**C** II only

**D** III only

**E** I and II only

**F** I and III only

**G** II and III only

**H** I, II and III

- 20** The line through  $(a, 0)$  and  $(0, b)$ , with  $a, b > 0$ , passes through  $(2, 3)$ . Which of the following statements is/are true?

**I**  $\frac{2}{a} + \frac{3}{b} = 1$

**II**  $a > 2$  and  $b > 3$

**III** The smallest possible value of  $ab$  is 24

**A** none of them

**B** I only

**C** II only

**D** III only

**E** I and II only

**F** I and III only

**G** II and III only

**H** I, II and III

**END OF TEST**